

Advanced Python for Data Analysis: Apple Stock Market Performance Report

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1. Dataset Overview

This project focused on analyzing historical Apple stock market data from June 2021 to June 2026 using Python. The dataset contained 1,255 trading-day records and included key stock market variables such as Open Price, High Price, Low Price, Close Price, and Trading Volume. The objective of the project was to apply advanced Python techniques for data transformation, time-series analysis, and feature engineering to uncover meaningful trends, market behavior, and investment insights.

2. Data Transformation Techniques Applied

Several data transformation and preprocessing techniques were applied using Pandas to prepare the dataset for analysis:

- Verified data quality through missing value, duplicate, and consistency checks.
- Validated chronological ordering of trading dates.
- Extracted time-based features including Year, Month, Quarter, and Day Name.
- Performed grouping and aggregation to analyze yearly and monthly performance trends.
- Restructured date information to support time-series analysis and visualizations.

The dataset was found to be clean, with no missing values, duplicate records, or data integrity issues.

3. Feature Engineering

To enhance the analysis, several new features were created:

- Trading Range (High Price – Low Price)
- Daily Price Change (Close Price – Open Price)
- Percentage Price Change
- Daily Return

- 7-Day Moving Average
- 30-Day Moving Average
- 30-Day Rolling Volatility
- Monthly Returns

These engineered features provided additional insight into stock performance, momentum, and market risk.

4. Trends Identified

The analysis revealed several notable trends:

- Stock prices followed a strong upward trajectory throughout the analysis period.
- Growth accelerated significantly after mid-2024, leading to new highs by 2026.
- Trading volume peaked in 2022 before gradually declining over subsequent years.
- Seasonal patterns were observed, with stronger returns occurring during specific months.
- Volatility increased during major market corrections before stabilizing.

5. Key Insights

- Stock prices increased from approximately 124 to over 315 during the analysis period, while average annual closing prices rose consistently from 146.74 to 270.46, demonstrating strong long-term growth.
- 2022 was the only year that recorded negative average daily returns (-0.097%), making it the weakest-performing period.
- July and November generated the strongest average daily returns, while September recorded the weakest performance.
- Stock prices continued rising despite declining trading volume after 2022, suggesting sustained investor confidence.
- The 7-Day and 30-Day Moving Averages confirmed persistent bullish momentum throughout most of the analysis period.

- The highest levels of market risk occurred during April–May 2025, when rolling volatility reached a peak of 4.41 before the stock recovered.

6. Recommendations

Based on the findings, the following recommendations are proposed:

- Monitor long-term trend indicators such as moving averages to identify sustained market direction.
- Pay close attention to periods of elevated volatility, as they often coincide with increased market uncertainty.
- Investigate major trading volume spikes to better understand the factors influencing investor behavior.
- Consider seasonal return patterns as supplementary indicators when evaluating market performance.
- Focus on long-term market trends rather than reacting to short-term price fluctuations.
- Balance growth opportunities with ongoing risk monitoring to support informed investment decisions.

7. Conclusion

The analysis demonstrated that Apple stock achieved strong long-term growth despite periods of market volatility and declining trading activity. While temporary corrections occurred throughout the period, the stock consistently recovered and maintained a positive long-term trend. Feature engineering, time-series analysis, and trend evaluation provided valuable insights into market behavior, investor confidence, and risk patterns. Overall, the project highlights the value of advanced Python techniques in transforming raw financial data into actionable insights for decision-making.

Appendix

Apple Stock Market Performance Dashboard

